

Things to Come

The inventor of hypertext forecasts the future, including the demise of the human race

Desktop computers

Very few computers will be on the desktop in 50 years. Today's workstation capacity will be in wristwatches, on the wall, off the wall, and so on.

Mobile computers

My favorite design, which I call the Walkie-Thinkie, is a wearable computer with a five-finger keyboard on your lapel (but there are lots of other options), a headphone, and a microphone. This device will have real-time digital recording and audio editing, a telephone, an answering machine, and voice mail built in. But I'm not holding my breath for reliable voice recognition.

User interfaces

It won't be the Mac/Windows-type interface or the voice interface seen in *Star Trek*. Both ideas are insane. The alternative is the design of abstract worlds and spaces with new rules and topologies that people can easily understand, use, and enjoy. Everybody should have the interface they want, so it will be increasingly important for "applications" to be delivered detached from "interfaces."

Operating systems

Linux will become very important. Then Taos, a very strong multitasking system for real time (i.e., time-compelling) applications, set-up boxes, and so on.

Virtual reality

That oxymoron for interactive realism will become just another interface and another big way to waste time. But it could also be very practical for lots of stuff—maps, instructions, and x-ray views.

On-line publishing

As I've always envisioned (and built into Project Xanadu), there will be automatic royalty on a per-ram basis for any portion of a document or object sold; stable publishing, so that materials may be reliably linked; unrestricted republication of any material, if that material uses Xanadu's "transcopyright" doctrine (i.e., a permission system that lets anyone reuse someone else's transcopyrighted material in a new document; you transmit a pointer to the desired material, and each customer purchasing your work buys each quotation from its original publisher).

Software-development tools

C++ will sink, C will resurge, Visual Basic (which ought



to be called Structured COBOL) will be big. Prograph, possibly the best, will catch on. CASE tools will become convenient.

Componentware

We need a king to make the rules about how the components divide and fit together. Standardization politics rules. But most of the issues of software will change completely, just as they have every five years.

Tomorrow's killer applications

My hope is that there will be no more applications, because future software will be set up to be tied together, piped together, scripted together. The real killer applications will be nanotech bugs that escape.

Players of the twenty-first century

Microsoft has peaked. By 2000, it should be second to Netscape. Silicon Graphics will live on. So will Sun. Thinking Machines will make a comeback. Apple will be acquired by Wal-Mart. Quarterdeck will make it big, but I'm not sure what with. IBM will die a terrible death, possibly bankruptcy or being acquired by Quarterdeck.

The data highway

It will be a conglomerate of parts on an Internet-like model, with some fast priority lanes and something like the Clipper chip for encryption—which everyone will bypass. But that doesn't matter, since it will let the government arrest whomever they choose, since we'll all be breaking the big, vague new law against everything.

New computing technology will change the world

In every possible way. But I find it hard to believe that the human race will survive to the year 2100. ■

Ted Nelson coined the concepts hypertext and hypermedia in 1965. He is best known for Project Xanadu. You can reach him on the Internet or BIX at edn@ccr.bcl.com.